Source Polarization and the Cross-Term Ledger
in a Finite Twin-Prime Signed-Gram Model

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Abstract

TPC-332 moved a five-control signed-Gram decomposition to a disjoint two-origin, three-scale ensemble, but left the arithmetic source norm as the live gate. This paper removes the dense operator and isolates the source polarization identity

$$\|\Lambda - b\|_2^2 = \|\Lambda\|_2^2 + \|b\|_2^2 - 2\langle \Lambda, b\rangle.$$

On the six parent-locked windows, the normalized cross-term coefficient $\kappa = 2\langle \Lambda, b\rangle/(\|\Lambda\|_2^2 + \|b\|_2^2)$ lies between 0.3548658992 and 0.3625023538. Thus the residual retains between 0.6374976462 and 0.6451341008 of the component-sum energy: the two components are neither nearly orthogonal nor nearly fully canceling on this finite panel. An independent reverse-factorization replay and mutation stress suite verify the ledger. This is a finite source diagnostic, not a source-uniform estimate, power saving, or twin-prime theorem.

1 Motivation and claim boundary

The session's dynamical twin-prime route uses a finite source vector of the form $\beta = \Lambda - b$ together with a signed prime-shell response. Earlier papers separated source norm from coordinate placement and then separated a control average from centered position response. The remaining arithmetic question is elementary in form but decisive in interpretation: how much of $\|\beta\|_2^2$ is explained by the cross term between the von Mangoldt-like component Λ and the comparison component b?

We make one deliberately narrow contribution. We certify the four terms of the polarization identity on a new finite source ensemble and test two extreme readings of the cross term:

1. "orthogonal": the cross term is negligible;
2. "complete cancellation": the cross term nearly equals the sum of the two component energies.

The finite data reject both readings only for the declared windows and model; the underlying quadratic-form expansion is standard (see, e.g., [1]). No statement below controls arbitrary origins, an unbounded cutoff, or the prime-pair counting function.

2 Finite source model

For $o \in \{42001, 44001\}$ and $N \in \{2048, 4096, 8192\}$, let

$$I_{o,N} = \{o, o+1, \ldots, o + N/2 - 1\}.$$

The source is the parent-locked finite model

$$\beta_o^{(2)}(t) = \Lambda(t+2) - b_o^{(2)}(t), \qquad b_o^{(2)}(t) = 2C_2\, \mathbf{1}_{2\nmid t} \prod_{\substack{p\mid t\\ p>2}} \frac{p-1}{p-2}. \tag{1}$$

Here $\Lambda(p^k) = \log p$ and is zero away from prime powers. The finite Euler product defining C_2 is evaluated through the inherited cutoff 50000, with the parent decimal midpoint and rational tail enclosure. All values $t+2$ in the six windows lie below this cutoff.

The symbols in (1) are therefore a declared finite numerical model, not an assertion that a truncated comparison is an exact global identity. The parent TPC-332 producer and certificate are locked by normalized SHA-256 in the machine-readable result.

3 Polarization ledger

For finite real vectors a, b, symmetry and bilinearity of the Euclidean inner product give

$$\|a-b\|_2^2 = \langle a-b, a-b\rangle = \|a\|_2^2 + \|b\|_2^2 - 2\langle a, b\rangle. \tag{2}$$

Applying this to $a = \Lambda$ and $b = b_o^{(2)}$ gives the source identity. Let

$$S = \|\Lambda\|_2^2 + \|b\|_2^2 > 0, qquad\kappa = \frac{2\langle \Lambda, b\rangle}{S}, qquad\rho = \frac{\|\Lambda - b\|_2^2}{S}.$$

Then, exactly,

$$\rho = 1 - \kappa, qquad\,\mathrm{corr}(\Lambda, b) = \frac{\langle \Lambda, b\rangle}{\|\Lambda\|_2\|b\|_2}. \tag{3}$$

The coefficient κ is a dimensionless diagnostic. It is not a positive-definite correlation coefficient and need not lie in $[0, 1]$ in an arbitrary model; the interval observed here is a finite result.

4 Protocol and exact anchor

The producer evaluates the source arrays for the six pairs (o, N), records the four terms in (2), and compares the two nested scales at each origin. It also stores nonzero-coordinate and coordinate-sign counts, so that a later support audit can use the same rows. The independent checker uses its own trial sieve, reverse distinct-factorization pass, and reverse order for the finite tail product. It does not import the producer's source routine. The stress program mutates the row count, a cross-term entry, the interval census, the firewall, and the exact anchor; every mutation is rejected.

The exact algebra anchor uses rational vectors

$$\Lambda = (3, -2, 5, 1), \qquad b = (1, 1, -1, 2), \qquad \Lambda - b = (2, -3, 6, -1).$$

It gives

$$\|\Lambda\|^2 = 39, \quad \|b\|^2 = 7, \quad \langle \Lambda, b\rangle = -2, \quad \|\Lambda - b\|^2 = 50 = 39 + 7 - 2(-2).$$

Reduced-fraction digests of these values are stored in the certificate. The anchor proves the finite algebra and does not pretend to be a prime-density experiment.

5 Results

Table 1 summarizes the six source windows. Every row falls in the predeclared diagnostic interval $(0.35, 0.37)$.

Table 1: Ranges over the six source windows.

quantity	minimum	maximum
$\kappa = 2\langle \Lambda, b\rangle / S$	0.35486589921455675	0.36250235375855522
$\rho = \|\Lambda - b\|^2 / S$	0.63749764624144467	0.64513410078544309
normalized cross-correlation	0.46455337638475735	0.48443427505641973

The largest recorded floating-point identity error is $1.4551915228366852 \times 10^{-11}$. The four adjacent-scale residual-energy growth factors, in origin order and scale order, are

$$1.8736551016394614, \quad 1.9695310092544431, \quad 1.9140068638900343, \quad 2.037675446375288.$$

The corresponding changes in κ are

$$-0.0047693401, \quad 0.0029351603, \quad -0.0016238452, \quad -0.0044484720.$$

These numbers describe finite-dimensional inclusions, not a convergence claim. In particular, the source cutoff is fixed while the vector dimension changes.

The observed coefficient has a useful interpretation. The cross term removes roughly 35–36 percent of the sum of component energies, leaving roughly 64 percent in the residual. This is enough to rule out a narrative in which the comparison term is irrelevant on this panel, but far from enough to establish a cancellation mechanism that could pay a power-saving gate.

6 What the ledger does and does not prove

The strongest positive result is a clean source-level interface: the four terms, dimensionless coefficient, and nested-scale comparisons are all recomputed independently on six disjoint-from-parent windows. The strongest obstruction is equally clear: finite source polarization is in a mixed regime, so neither near orthogonality nor near total cancellation is a safe replacement for a future uniform estimate.

The release labels are:

- `PROVED_EXACT_FINITE`: (2) and (3);
- `NUMERICALLY_CERTIFIED_FINITE`: six rows, four scale pairs, independent replay, and five mutation rejections;
- `REFUTED_SCOPED`: the two extreme cross-term readings on this finite panel;
- `OPEN`: a source-uniform arithmetic L^2 bound, support attribution, strict 1/400 payment, Route-B Gate B, and a twin-prime conclusion.

Thus

`ARITHMETIC_ADVANCE=NO`, `FIXED_POWER_CREDIT=0`, `FULL_GATE_B=OPEN`, `TWIN_PRIME_RESULT=NONE`.

The Session-named Route-A and Route-B evaluator files are not present in this checkout. The local Bridge-B check is consequently a fail-closed repository test, not an official evaluator pass.

7 Next question

The cross term is substantial, but its numerical value alone does not say which arithmetic coordinates create it. The next natural paper will split $\langle \Lambda, b \rangle$ into actual twin-prime coordinates, prime-power coordinates, and the odd composite background. That support ledger is the smallest test of whether the source polarization has any twin-prime-specific content.

References

[1] Roger A. Horn and Charles R. Johnson. *Matrix Analysis*. Cambridge University Press, 2 edition, 2013.